

Curriculum Vitae

PERSONAL INFORMATION

Hongyu Li
Brown University
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GitHub: <https://github.com/lhy0807>
Google Scholar: <https://scholar.google.com/citations?user=aM2PHREAAAAJ&hl=en>

EDUCATION

- 08/2023–Current **Ph.D., Computer Science**
Brown University, Providence, RI, USA
- 08/2021–05/2023 **M.Sc., Computer Science**
Northeastern University, Boston, MA, USA
Thesis: Toward Stereo-based Obstacle Detection using Efficient Deep Neural Network
- 08/2018–12/2020 **B.Sc., Computer Science**
Rensselaer Polytechnic Institute, Troy, NY, USA
Second Major: Economics
Honors: Cum Laude, Dean's Honor List

EXPERIENCE

- 06/2025–TBD **Research Intern**
 Robotics and AI Institute (fka Boston Dynamics AI Institute), Cambridge, MA
Hosts: [Jiahui Fu](#).
Perception research.
- 05/2024–09/2024 **Applied Scientist II Intern**
 Amazon Fulfillment Technologies & Robotics (Innovation Lab), Westborough, MA
Hosts: Professor [Taşkın Padır](#) and [Tye Brady](#).

Researched in-hand object 6D pose tracking using visuotactile perception. Published paper [C.2](#).

05/2023–08/2023 **Research Intern**

🏢 Honda Research Institute, San Jose, CA

Hosts: Dr. Nawid Jamali and Dr. Soshi Iba.

Researched visuotactile perception. Published paper [C.5](#). Filed patents [P.1](#) and [P.2](#).

09/2022–12/2022 **Research Intern**

🏢 Honda Research Institute, San Jose, CA

Hosts: Dr. Nawid Jamali and Dr. Soshi Iba.

Researched in-hand object 6D pose estimation using visuotactile perception. Published paper [J.2](#). Filed patents [P.3](#) and [P.4](#).

09/2021–05/2023 **Graduate Research Assistant**

Robotics and Intelligent Vehicles Research Lab, Northeastern University, Boston, MA

PI: Professor Taşkın Padır

Researched in the fields of robot perception and navigation. Published papers [J.1](#), [C.6](#), and [C.7](#).

03/2021–06/2021 **Artificial Intelligence Intern**

🏢 KPMG, Nanjing, China

Developed the speech processing modules for the KPMG AI Factory Platform.

PUBLICATIONS

* or [†] represents equal contribution or equal advising, depending on the roles of the authors.

Journals

J.1 Linfeng Zhao*, **Hongyu Li***, Taşkın Padır, Huaizu Jiang[†], Lawson L.S Wong[†]. *E(2)-Equivariant Graph Planning for Navigation*. Accepted by IEEE Robotics and Automation Letters (RA-L) 2024. Present at IROS 2024.

J.2 **Hongyu Li**, Snehal Dikhale, Soshi Iba, Nawid Jamali. *ViHOPE: Visuotactile In-Hand Object 6D Pose Estimation with Shape Completion*. Accepted by IEEE Robotics and Automation Letters (RA-L) 2023. Present at ICRA 2024.

Conferences

C.1 **Hongyu Li**, Mingxi Jia, Mete Tuluhan Akbulut, Yu Xiang, George Konidakis, Srinath Sridhar. *V-HOP: Visuo-Haptic 6D Object Pose Tracking*. Accepted by Robotics: Science and System (RSS) 2025.

- C.2 Hongyu Li**, James Akl, Srinath Sridhar, Tye Brady, Taşkın Padır. *ViTa-Zero: Zero-shot Visuotactile Object 6D Pose Estimation*. Accepted by IEEE International Conference on Robotics and Automation (ICRA) 2025.
- C.3 Hongyu Li**, Taşkın Padır, Huaizu Jiang. *StereoNavNet: Learning to Navigate using Stereo Cameras with Auxiliary Occupancy Voxels*. Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024.
- C.4 Tianye Ding***, **Hongyu Li***, Huaizu Jiang. *ODTFormer: Efficient Obstacle Detection and Tracking with Stereo Cameras Based on Transformer*. Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024.
- C.5 Hongyu Li**, Snehal Dikhale, Jinda Cui, Soshi Iba, Nawid Jamali. *HyperTixel: Hyper-Resolution for Tixel-Based Tactile Signal Through Contrastive Learning*. Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024.
- C.6 Hongyu Li**, Zhengang Li*, Neşet Ünver Akmandor*, Huaizu Jiang, Yanzhi Wang, Taşkın Padır. *StereoVoxelNet: Real-Time Obstacle Detection Based on Occupancy Voxels From a Stereo Camera Using Deep Neural Networks*. Accepted by IEEE International Conference on Robotics and Automation (ICRA) 2023. Project homepage: <https://lhy.xyz/stereovoxelnet>
- C.7 Neşet Ünver Akmandor**, **Hongyu Li**, Gary M. Lvov, Eric Dusel, Taşkın Padır. *Deep Reinforcement Learning based Robot Navigation in Dynamic Environments using Occupancy Values of Motion Primitives*. Accepted by IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022.

PRESENTATIONS

1. **Hongyu Li**, Snehal Dikhale, Soshi Iba, Nawid Jamali. *ViHOPE: Visuotactile In-Hand Object 6D Pose Estimation with Shape Completion*. Spotlight Talk at [NeurIPS 2023 Workshop on Touch Processing](#).
2. **Hongyu Li**, Huaizu Jiang, Taşkın Padır. *StereoNavNet: Learning to Navigate using Stereo Camera with Auxiliary Occupancy Voxels*. Spotlight Talk at CVPR 2023 Workshop on 3D Vision and Robotics.
3. **Hongyu Li**, Zhengang Li*, Neşet Ünver Akmandor*, Huaizu Jiang, Yanzhi Wang, Taşkın Padır. *StereoVoxelNet: Real-Time Obstacle Detection Based on Occupancy Voxels From a Stereo Camera Using Deep Neural Networks*. Presented at IROS 2022 workshop "Agile Robotics: Perception, Learning, Planning, and Control," Kyoto, Japan, October 2022.

AWARDS

1. ICRA 2023 Travel Grant

PATENTS

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- P.1 Hongyu Li**, Nawid Jamali, Soshi Iba. SYSTEMS AND METHODS FOR TAXEL HYPER-RESOLUTION THROUGH MULTI-CONTACT LOCALIZATION.  *Patent Pending*.
- P.2 Hongyu Li**, Nawid Jamali, Soshi Iba. REPRESENTATION LEARNING OF TAXEL-BASED TACTILE SENSOR.  *Patent Pending*.
- P.3 Hongyu Li**, Nawid Jamali, Snehal Dikhale, Soshi Iba. SYSTEMS AND METHODS FOR VISUOTACTILE OBJECT POSE ESTIMATION WITH SHAPE COMPLETION. US-20240371022-A1.  *Patent Pending*.
- P.4 Hongyu Li**, Nawid Jamali, Soshi Iba. SYSTEMS AND METHODS FOR A SHAPE COMPLETION MODEL.  *Patent Pending*.
- P.5** Feiyu Zhu, Tianjian Dai, **Hongyu Li**, Tiancheng Mai, Yuchen Liang, Xiaojing Su. A kind of ball serving device for balls sport training. Granted CN107007996A

TEACHING

- 01/2024–05/2024 **Graduate Teaching Assistant**
Department of Computer Science, Brown University, Providence, RI
CSCI 1430 Introduction to Computer Vision
- 09/2021–05/2022 **Graduate Teaching Assistant**
Khoury College of Computer Sciences, Northeastern University, Boston, MA
CS 7610 Foundations of Distributed Systems
DS 2500 Intermediate Programming with Data
- 09/2019–07/2020 **Teaching Assistant**
Department of Computer Science, Rensselaer Polytechnic Institute, Troy, NY
CSCI 1190 Beginning Programming for Engineers

SERVICES

Reviewer

Conferences

1. IEEE International Conference on Robotics and Automation (**ICRA**) 2024, 2025
2. IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**) 2024, 2025
3. IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) 2024, 2025
4. European Conference on Computer Vision (**ECCV**) 2024
5. International Conference on Computer Vision (**ICCV**) 2025
6. Annual AAAI Conference on Artificial Intelligence (**AAAI**) 2025
7. ACM CHI Conference on Human Factors in Computing Systems (**CHI**) 2024
8. IEEE-RAS International Conference on Humanoid Robots (**Humanoids**) 2024

Journals

9. IEEE Robotics and Automation Letters (**RA-L**)
10. IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**)
11. Elsevier **Neurocomputing**

Providence, RI, May 2, 2025